# An Analysis of Population Structure of Arctic Fox in Greenland and Arctic Circle Area

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## Abstract

This study investigates the population structure of the Arctic fox (*Vulpes lagopus*) in Greenland and the circumpolar region, focusing on the role of sea-ice connectivity in shaping gene flow. We analyzed a high-density genomic dataset of 0.8 million SNPs from Greenland, Canada, and Siberia. Genetic structures were characterized using PCA and ADMIXTURE clustering following quality control and LD pruning via PLINK, with evalAdmix ensuring the robustness of ancestral inferences. Genetic differentiation and diversity were further quantified using pairwise Fst and runs of homozygosity (ROH) analysis. The results reveal a distinct geographical structure among Greenlandic populations, with evidence of significant sea-ice mediated connectivity. PCA identified three explicit immigrants in Scoresbysund whose genomic profiles align with distant circumpolar metapopulation. Notably, several individuals from Zackenberg and Kangerlussuaq exhibited a genetic inclination toward the circumpolar metapopulation in the PCA. ADMIXTURE analysis confirmed that these individuals harbor both local and external ancestral components, suggesting they are admixed descendants of historical migrants rather than primary immigrants. These findings demonstrate that sea-ice bridges facilitate not only long-distance individual dispersal but also substantial inter-population gene flow and genetic introgression.

**Keywords:** Arctic fox; Population structure; Sea-ice connectivity; Gene flow; Genetic admixture

## Introduction

The arctic fox (*Vulpes lagopus*) is a small-sized carnivore which is adapted to arctic environments, and is characterized by short morphology features that minimize heat loss. It is a circumpolar species and currently inhabits tundra regions across Greenland, North America, Eurasia, Iceland. Arctic fox populations are relatively abundant, although there are some fluctuations relating to prey availability, particularly lemmings, which represent the species' main food source[1].

Arctic foxes exhibit a remarkable movement ecology, with individuals being able to travel very long distances, across sea ice and terrestrial environments. Satellite telemetry studies have shown that both juveniles and adults disperse across large Arctic areas, often covering total distances of several thousand kilometers. This highlights the species' exceptional dispersal capacity[2]. These long-distance movements are facilitated by the presence of sea ice, which allows Arctic foxes to inhabit distant areas. Sea ice acts as a bridge, connecting far flung tundra habitats and distant populations across the circumpolar distribution of the species. Arctic foxes seasonally move over the sea ice, to find food sources. These movements are mostly connected to lemming population cycles, and can lead them to new areas, influencing the gene flow[3].

Greenland is a unique and important system for studying Arctic fox population structure. This dynamic can be largely attributed to Greenland's unique biogeographical paradigm. While Greenland functions as a strictly isolated landmass separated by expansive marine barriers during the warmer months, the extensive formation of winter sea ice establishes ephemeral terrestrial corridors linking it to both the North American and Eurasian continents. For highly mobile Arctic species, these seasonal cryospheric bridges serve as critical dispersal conduits. They facilitate episodic, long-distance migration events

across the Holarctic region, thereby enabling periodic gene flow. Such intermittent genetic admixture plays a pivotal evolutionary role, effectively mitigating the loss of heterozygosity and counteracting the effects of genetic drift that would otherwise dominate in geographically restricted insular populations[3,4]. This combination of isolation and connectivity allows both regional differentiation and population mixing, making Greenland a very unique and ideal study area for understanding the interplay between movement ecology and genetic structure.

Genetic studies have shown that populations connected by sea ice generally have low genetic differentiation, indicating extensive gene flow. In contrast, populations isolated from sea ice, such as those in Iceland, and parts of Scandinavia, exhibit higher genetic divergence. These findings demonstrate the important role of sea ice connectivity in creating population genetic structures[3,4]. Mitochondrial analyses also revealed that many haplotypes are shared across distant, ice-connected regions, including North America, northern Greenland, Svalbard, Siberia, reflecting an extensive historical gene flow[5]. Recent microsatellite studies further showed that even among connected populations, small local differences exist because related individuals tend to stay close together[6]. In addition, Arctic foxes can be divided into two ecotypes, lemming and coastal, based on their primary food sources. This food preference leads to habitat imprinting, which means that foxes prefer to inhabit environments similar to their natal habitat, creating genetic subdivisions even without physical barriers[7]. Together, these studies show the importance of interactions among movement ecology, sea-ice connectivity, and historical population processes in shaping Arctic foxes' genetic structure.

In this study, our goal is to investigate the populational structure of arctic foxes in Greenland, focusing on how movement patterns and sea-ice connectivity have influenced gene flow among populations. We analyzed samples from Zackenberg(n=5), Scoresbysund(n=12), Qanisartuut(n=11), and Kangerlussuaq(n=10), and included also samples from Canada and Siberia(n=9) which contain ~0.8 million bi-allelic SNPs to understand both local and regional patterns of connectivity (Fig 1).

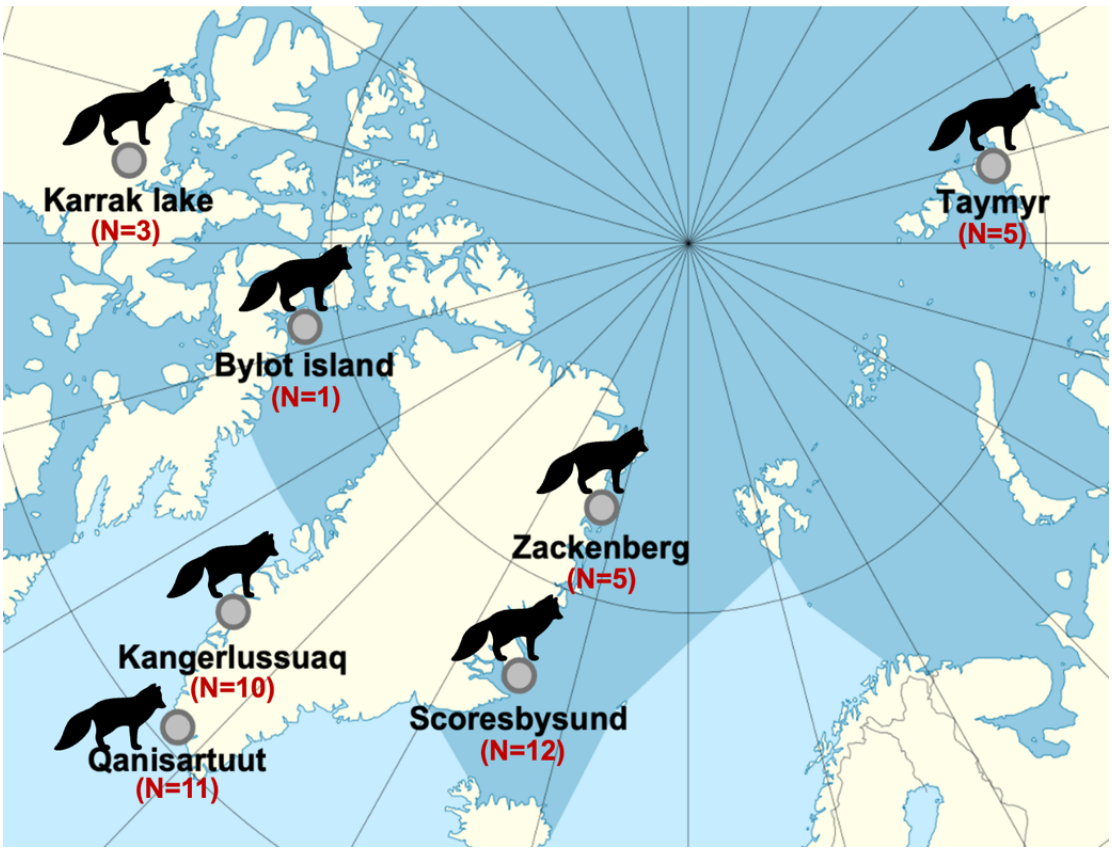


**Fig 1. Geographical distribution of the study sites across the Arctic region.** This map illustrates the locations and samples numbers of the seven regions in this study: Karrak Lake, Bylot Island, Kangerlussuaq, Qanisartuut, Scoresbysund, Zackenberg, and Taymyr.

# Methods

### Study Species and Genomic Dataset

The genomic data in this study consisted of an imputed and thinned SNP dataset of the Arctic fox (*Vulpes lagopus*), provided by the Population Genetics Course at the University of Copenhagen. To ensure computational efficiency while maintaining genome-wide representation, the original dataset was subsampled to a high-quality subset of markers. These sample data are sampled from Greenland (Kangerlussuaq, Qanisartuut, Scoresbysund, Zackenberg), Canada (Bylot island, Karrak lake) and Siberia (Taymyr).

### Genotype Quality Control and PCA

Genotype quality control and Principal Component Analysis (PCA) were performed using PLINK v1.9[8]. SNPs were filtered based on missingness (--geno 0) and minor allele frequency (--maf 0.05) during the initial data processing stage. To ensure marker independence and minimize redundancy, linkage disequilibrium (LD) pruning was applied ($r^2 < 0.1$, window size 50, step 10). The resulting set of independent SNPs was then used for PCA to characterize the population structure.

### Population Structure and Admixture Analysis

Population genetic structure was inferred using ADMIXTURE (v1.3.0)[9] based on the pruned SNP dataset. The optimal number of ancestral clusters was evaluated by testing K values ranging from 2 to 7. Stability and convergence of the clustering were ensured by conducting 100 independent iterations for each K using distinct random seeds. The resulting ancestry proportions (Q-matrices) and allele frequencies (P-matrices) were recorded for each run to facilitate downstream cross-validation and cluster alignment. For each tested K, the run with the highest log-likelihood was selected. The evalAdmix (v0.95)[10] was employed to evaluate inter-individual residual correlations, ensuring the reliability of the inferred population structure. A high correlation of residuals indicates a potential underfitting or over-specification of K. This step ensured that the chosen ancestral clusters accurately represented the underlying genetic variation without significant systematic bias.

### Genetic Differentiation

Genetic differentiation between geographical regions was quantified using pairwise $F_{ST}$ estimators. Calculations were performed following the Weir and Cockerham (1984) method[11], implemented via a custom R script provided in exercise in our course. This approach accounts for varying sample sizes and provides a weighted average across all thinned, non-missing SNP loci. All possible pairwise comparisons between identified regions were conducted, and the resulting $F_{ST}$ values were used to assess the degree of genetic isolation among Arctic fox populations.

### Heterozygosity and Runs of Homozygosity

Individual observed heterozygosity and runs of homozygosity (ROH) were aggregated by region. Non-parametric methods were employed due to the non-normal nature of the data: the Kruskal–Wallis test was used for global comparisons among regions, followed by Wilcoxon rank-sum tests for pairwise comparisons. All P-values were corrected for multiple testing.

## Results

Principal Component Analysis (PCA) revealed a distinct genetic structure among the four Greenlandic Arctic fox populations, with most individuals forming region-specific clusters. However, distant groups from Canada and Siberia tend to form cluster. Notably, three individuals (ArcFoxSamp_128, ArcFoxSamp_144, ArcFoxSamp_152) from the Scoresbysund region (colored in orange) did not cluster with their geographical neighbors. Instead, these outliers exhibited strong genetic similarity with extrinsic populations from Canada and Siberia, positioning them far from the main Greenlandic cluster and within the genetic space defined by the circumpolar outgroups.

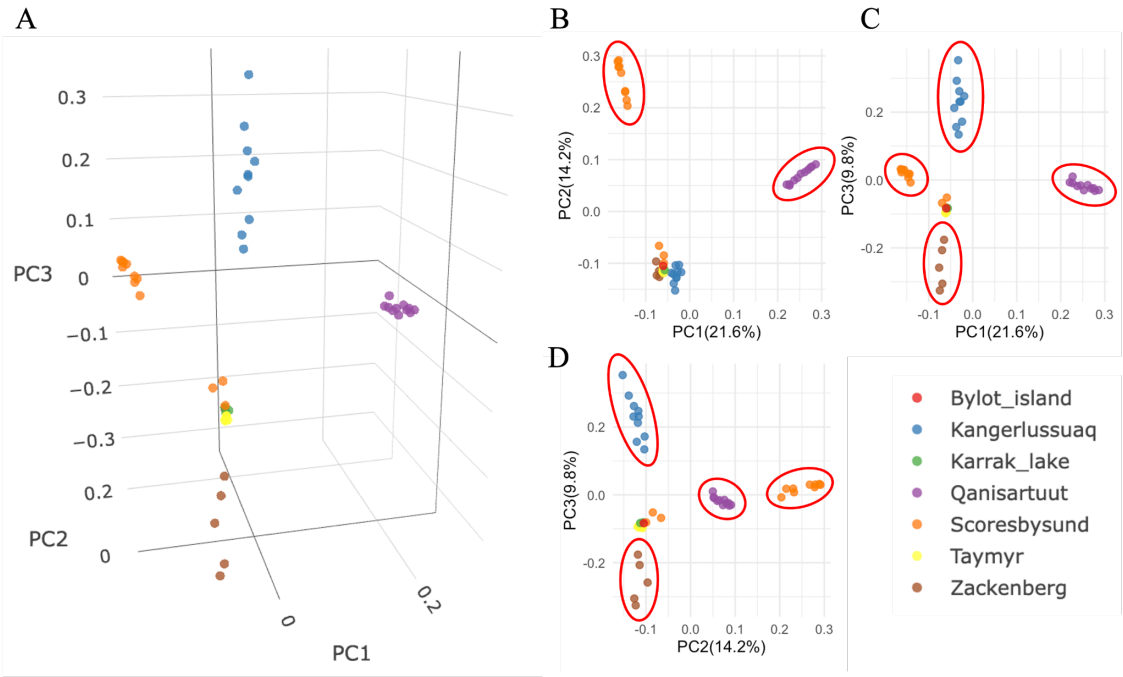


**Fig 2. Principal Component Analysis (PCA) of Arctic fox populations.** Scatter plots show the first three principal components (PCs) explaining the major genetic variation across individuals. (Left: PC1 vs PC2; Center: PC1 vs PC3; Right: PC2 vs PC3). Individuals are color-coded by sampling region. Red circle indicates these potential clusters of subpopulations.

To determine the most representative number of ancestral clusters (K), we performed ADMIXTURE analysis for K=2 to 7, with 100 independent iterations per K to ensure global optimization. For each K value, the iteration with the highest log-likelihood was identified as the best-fit model: specifically, Run 27 for K=2, Run 68 for K=3, Run 93 for K=4, Run 63 for K=5, Run 25 for K=6, and Run 78 for K=7 .The robustness of these top-ranking models was further validated using evalAdmix. The analysis of residual correlations between individuals revealed that K=5 provided the most stable and biologically parsimonious fit to our data, with the lowest systematic bias compared to other values. Consequently, the results from K=5 (Run 63) were selected for the final characterization of the Arctic fox population structure, as this model best captured the genetic differentiation between the Greenlandic regions and the circumpolar outgroups.

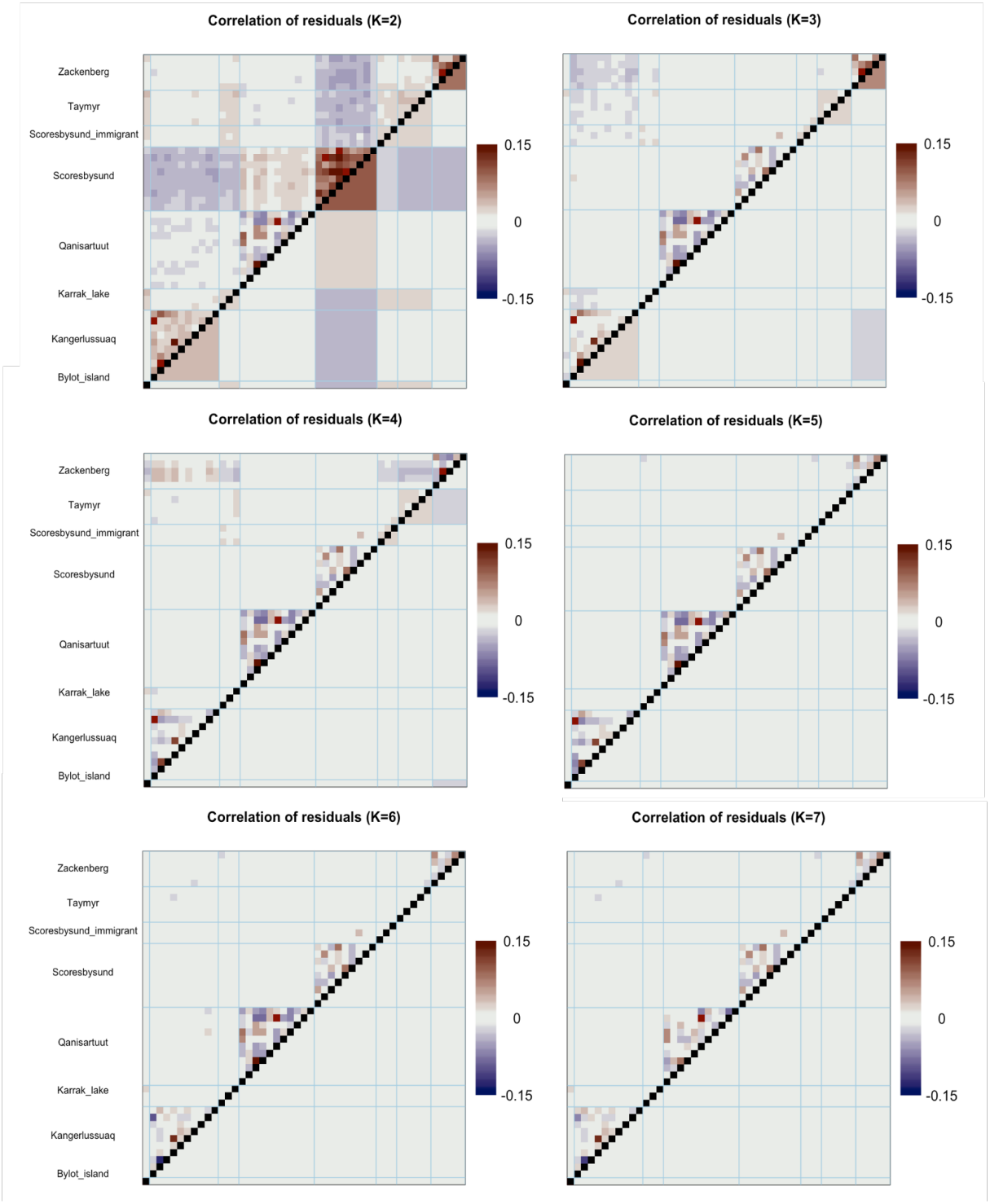


**Fig 3. Evaluation of ADMIXTURE model fit using residual correlation matrices for K=2 to 7.** The heatmaps display the correlation of residuals between individuals across different K values. Darker colors (brown/blue) indicate systematic deviations where the model fails to explain the genetic data (under-fitting). At K=5, the matrix shows a predominantly uniform light-grey distribution, indicating that the five-cluster model effectively captures the underlying population structure with minimal residual correlation. This validates K=5 as the most parsimonious and robust model for the dataset.

The ADMIXTURE analysis at K=5, identified as the most parsimonious model via evalAdmix residual correlations, revealed a pronounced genetic structure among Greenlandic Arctic fox populations. The four primary regions Qanisartuut (QAN), Scoresbysund (SCO), Zackenberg (ZAC), and Kangerlussuaq (KAN) exhibited high levels of genetic independence, with most individuals assigned to a single, region-specific ancestral component. This suggests early divergence and minimal recent gene flow

between these established populations. However, striking genetic outliers were identified within the northern groups. Specifically, three individuals from Scoresbysund (SCIM) displayed an ancestry profile dominated by a "green" component, indicating high genetic affinity with Canadian (Bylot Island and Karrak Lake) and Siberian (Taymyr) outgroups rather than their geographical neighbors. Similarly, individuals ArcFoxSamp_104, 183, and 191 in the Zackenberg population exhibited substantial green and blue components, suggesting they represent non-local lineages. While the Kangerlussuaq population also displayed an admixed profile—characterized by a dominant red component alongside a persistent green ancestral signature—this subtle admixture was less discernible in the PCA. Conversely, the ADMIXTURE results for Zackenberg are explicitly corroborated by the PCA, where several Zackenberg individuals (represented by brown data points) cluster in close proximity to the circumpolar outgroup, further validating their status as potential historical migrants.

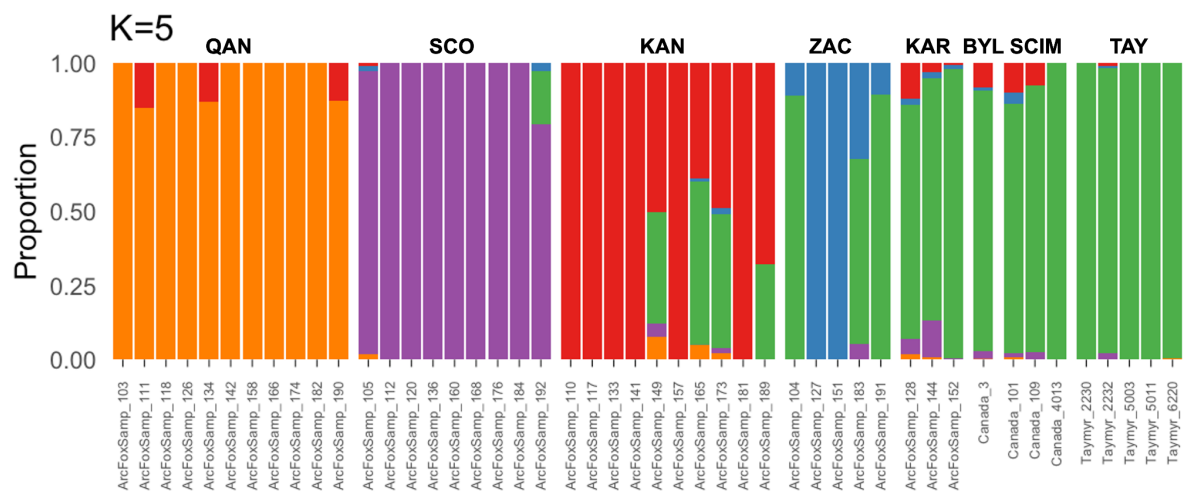


**Fig 4. Genetic structure of Arctic fox populations inferred by ADMIXTURE at K=5.** Each vertical bar represents an individual, and the colors represent the proportional assignment to five ancestral clusters. Individuals are grouped by geographic region: QAN: Qanisartuut; SCO: Scoresbysund; KAN: Kangerlussuaq; ZAC: Zackenberg; KAR: Karrak Lake; BYL: Bylot Island; SCIM: Scoresbysund Immigrants; TAY: Taymyr.

The pairwise $F_{ST}$ analysis, calculated using the Weir and Cockerham (1984) estimator, reveals a complex population structure for the Arctic fox, characterized by both high connectivity across vast distances and localized genetic isolation. A striking feature of the genomic landscape is the near-zero differentiation ($F_{ST} \approx 0.000 - 0.001$) between the Siberian (Taymyr) and Canadian (Bylot Island, Karrak Lake) populations. Despite the thousands of kilometers separating these regions, the negligible $F_{ST}$ values suggest an extremely high level of gene flow, likely facilitated by the species' use of seasonal sea ice as a corridor for long-distance dispersal. In contrast, populations in Greenland exhibit significantly higher levels of genetic structure; Qanisartuut, in particular, shows the highest degree of isolation with $F_{ST}$ values ranging from 0.087 to 0.141 against all other groups. This isolation is probably driven by the more sedentary "coastal" ecotype of foxes in Southern Greenland, which rely less on sea-ice-driven migrations than their high-arctic counterparts. Furthermore, the inclusion of the "Scoresbysund_immigrant" provides a compelling case study for real-time gene flow; although sampled in Scoresbysund, this individual is genetically indistinguishable from the Taymyr and Canadian populations, highlighting the stark contrast between resident lineages and transient migrants. These relationships are visually summarized by the hierarchical clustering dendrogram, which bifurcates the study sites into a divergent Greenlandic cluster and a highly mobile, panmictic-like circumpolar cluster.

**Table 1. Summary of Genetic Differentiation Patterns**

| Comparison | $F_{ST}$ Range | Interpretation |
|---|---|---|
| Taymyr vs. Canada | 0.000 – 0.001 | High Gene Flow / Panmixia |
| Qanisartuut vs. All | 0.087 – 0.141 | Significant Genetic Isolation |
| Scoresbysund vs. Immigrant | 0.059 | Resident/Migrant Distinction |

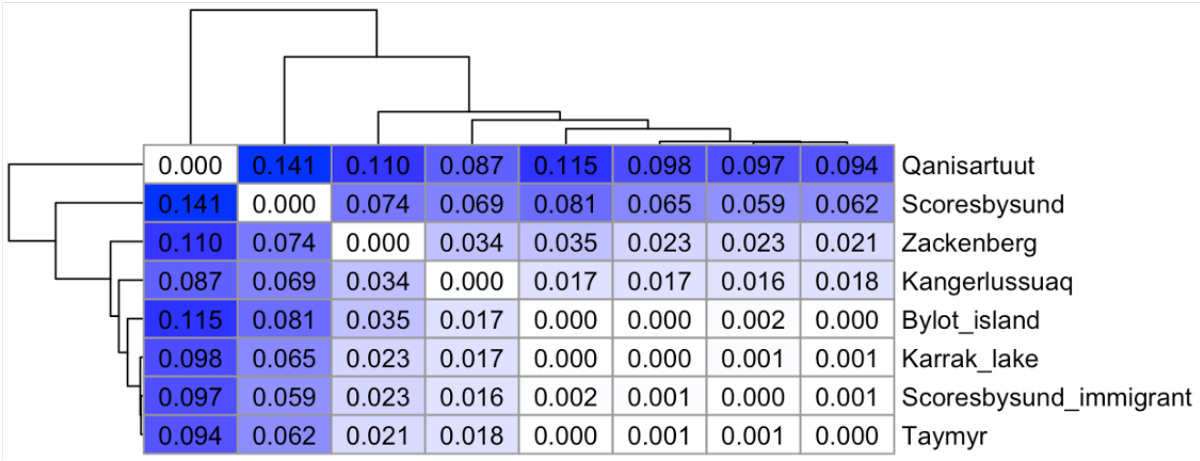


**Fig 5. $F_{ST}$ results among Arctic fox populations.** A heatmap of pairwise $F_{ST}$ values. The color intensity represents the degree of genetic differentiation, with darker blue indicating higher $F_{ST}$ values and greater genetic isolation. The dendrogram (top and left) illustrates the hierarchical clustering of populations based on genetic similarity.

The mean observed heterozygosity varied among different regions (Fig 6). Qanisartuut had the lowest mean $H_0$ and Canada and Taymyr had the highest values. Within the different regions, there was variation, particularly in Qanisartuut, likely influenced by individuals with lower heterozygosity values.

Total runs of homozygosity (ROH) length also varied among regions (Fig 7). Individuals from Qanisartuut showed the highest total ROH length, followed by Scoresbysund and Kangerlussuaq, while Canada and Taymyr showed the lowest ROH values. Zackenberg and Scoresbysund_immigrant showed relatively low ROH compared to other Greenland regions. Overall, ROH length differed among regions, indicating differences in levels of homozygosity among Arctic fox populations.

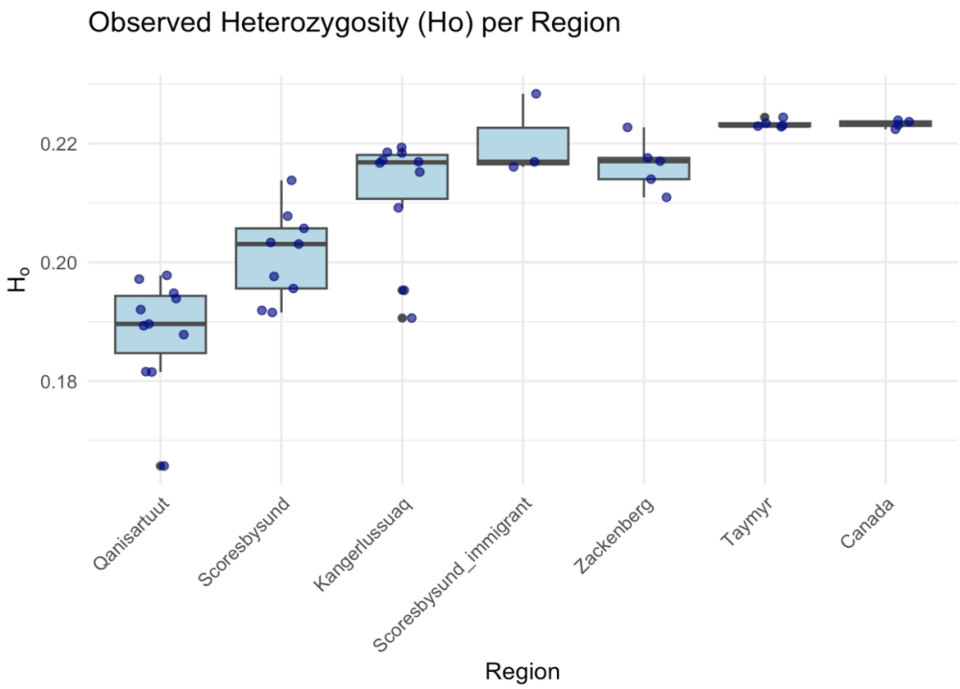


**Fig 6. Observed heterozygosity (Ho) per region.** Each point represents an individual, and boxplots summarize heterozygosity distribution within each region.

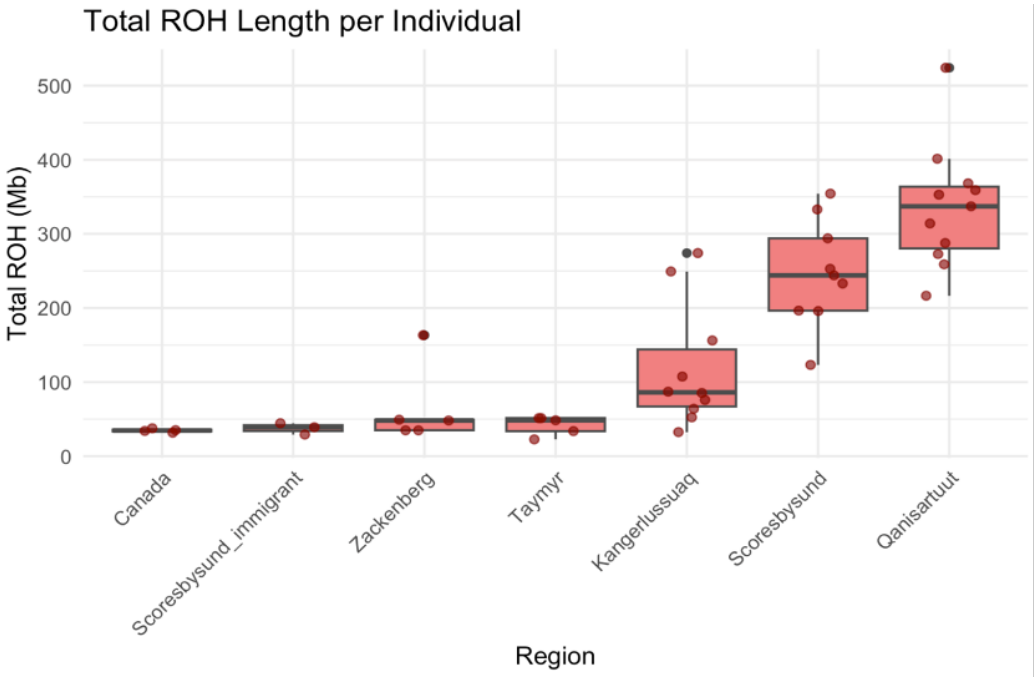


**Fig 7. Total runs of homozygosity (ROH) length per individual across regions.** Each point represents an individual, and boxplots summarize the distribution of ROH length within each region.

## Discussion

The genetic affinities observed between Siberia, Canada, and the immigrant individuals in Greenland suggest the existence of a vast, interconnected circumpolar meta-population. In ADMIXTURE result, we propose that the green and orange ancestral components represent the genomic signatures of this broad meta-population, facilitated by the Arctic fox's remarkable capacity for long-distance dispersal across sea ice. The presence of the local purple component in the Scoresbysund immigrants indicates that these individuals have initiated genetic exchange with the resident population upon arrival, demonstrating active gene flow following dispersal. The genetic composition of Kangerlussuaq suggests a historical expansion model where an ancestral northward-moving group interacted with the circumpolar meta-population; over time, the ancestral green component likely diminished through lineage sorting and genetic drift, while the localized red component became dominant. This pattern is less detectable in PCA space because the overall genetic drift in Kangerlussuaq has pulled the population toward its own cluster, yet the information of its ancestral connection remains visible in the ADMIXTURE proportions.

The stark contrast in isolation between the southern Qanisartuut population and the rest of regions in Greenland is primarily driven by sea ice dynamics. In northern Greenland, winter sea ice provides transient bridges that enable a "stepping-stone" dispersal model across the Arctic. The Scoresbysund region, in particular, appears to function as an evolutionary "sink" or secondary contact zone, where extralimital lineages colonize the island during periods of high sea-ice connectivity and become sequestered during subsequent isolation phases. This episodic connectivity explains why high-latitude populations maintain such high genetic similarity across vast geographical distances while simultaneously allowing for the development of distinct regional clusters within Greenland.

Previous microsatellite and mtDNA studies have reported significant genetic differentiation among Greenlandic Arctic fox localities such as Scoresbysund, Qaqortoq, and Kangerlussuaq[12]. This is consistent with our SNP-based evidence of strong structure in Greenland, particularly the distinctiveness of South Greenland (Qanisartuut) and moderate differentiation at Scoresbysund.

Observed heterozygosity and runs of homozygosity analyses showed variation among populations. Because our dataset includes only 47 individuals in total, the results might be affected. Qanisartuut showed the lowest heterozygosity and the highest total ROH, including some clear outliers. This could be consistent with reduced diversity, but we cannot make strong conclusions about a bottleneck or founder effect from these measures alone.

We also tested for a bottleneck or founder effect in Greenlandic Arctic foxes using π, Watterson's θ, Tajima's D, and the folded SFS (Fig 8, A-D). Although Tajima's D is negative in all populations, the values are extremely negative and likely reflect artifacts of imputation, SNP thinning, and small sample size rather than clean demographic signal. The folded SFS is also low-resolution because downsampling is constrained by the smallest sample, and folding further reduces interpretability. Diversity statistics do not show a strong, consistent reduction in Greenland relative to source populations. Therefore, our summary statistics do not reveal a clear bottleneck or founder-effect signal in Greenland, but the data are not well suited for robust demographic inference, so a bottleneck cannot be excluded.

Several limitations of this study warrant consideration. Notably, the stringent data filtering applied may have obscured subtle signals necessary for deeper analyses of population history and gene flow. Future research should integrate datasets from a broader range of sources to enable higher-resolution reconstructions of the origins and temporal dynamics of the identified migration events. Furthermore, quantifying the genetic divergence between the Qanisartuut and Meta populations could facilitate the estimation of historical ice bridge formation periods, thereby offering novel insights into paleoclimatology and biogeography.

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## Supplementary Information

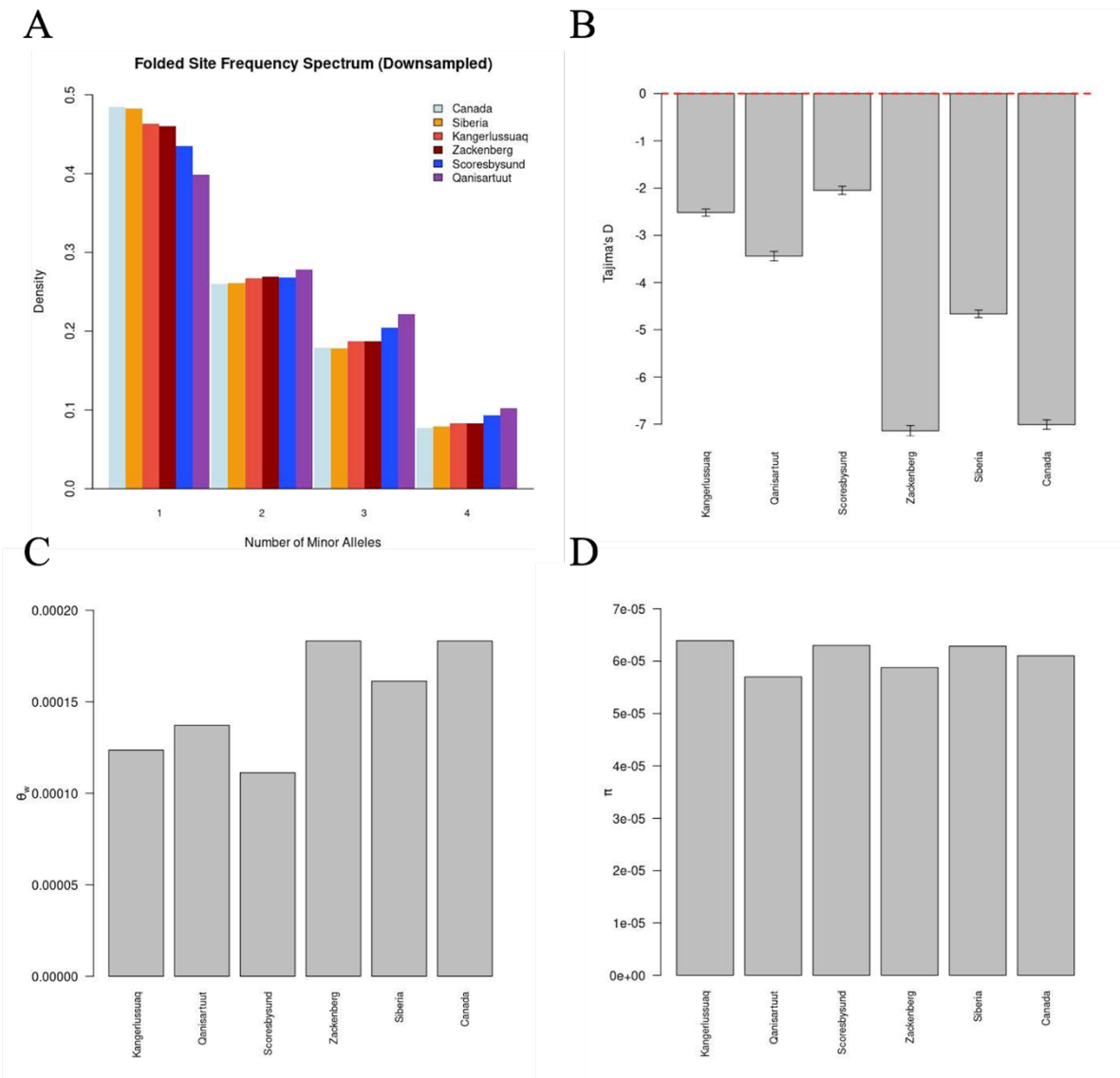


**Fig 8. Barplot results of Folded SFS, Tajima'D, Watterson's θ, and π**